

JEE Main - 18 | JEE - 2022

Date: 23/01/2022

Maximum Marks: 360

Timing: 09:00 AM to 09:00 PM

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **360**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **10 Numerical Value Type Questions Out of which ONLY 5 questions have to be attempted**.
The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

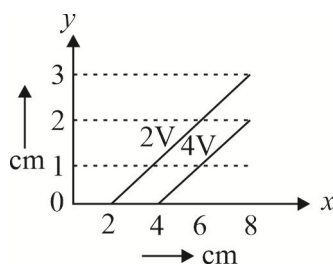
PART - I : PHYSICS

120 MARKS

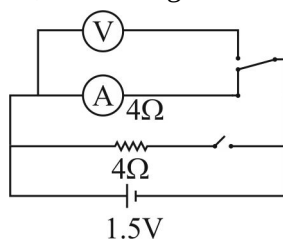
SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

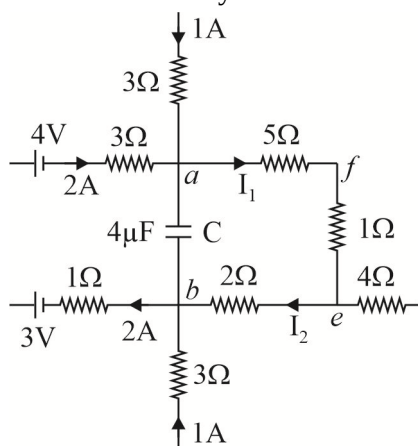
1. Figure shows two equipotential lines in x, y plane for an electric field. The x component (E_x) and y component (E_y) of field in space between these lines are respectively:



- (A) $-100\text{Vm}^{-1}, 200\text{Vm}^{-1}$ (B) $-100\text{Vm}^{-1}, -200\text{Vm}^{-1}$
 (C) $200\text{Vm}^{-1}, 100\text{Vm}^{-1}$ (D) $100\text{Vm}^{-1}, 100\text{Vm}^{-1}$
2. When terminals of a cell of emf 1.5V is connected to an ammeter of resistance 4Ω , the reading is 0.3A. The reading of ammeter when a 4Ω resistor is connected between terminals of cell is A. If ammeter is replaced by a high resistance voltmeter, the reading is V. Then A and V are :

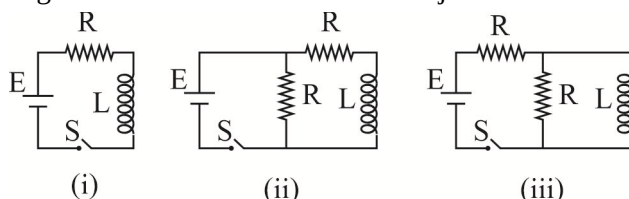


- (A) 0.25A, 1.25V (B) 5.2A, 2.1V (C) 0.25A, 1.2V (D) 2.1A, 5.2V
3. A part of a circuit at steady state is shown. The energy stored in capacitor C in steady state is :

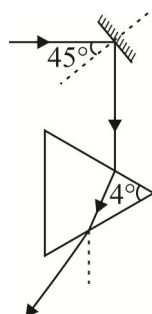


- (A) $8 \times 10^{-4} \text{ J}$ (B) $10 \times 10^{-4} \text{ J}$ (C) $6 \times 10^{-4} \text{ J}$ (D) $12 \times 10^{-4} \text{ J}$
4. The true value of angle of dip at a place is 60° , the apparent dip in a plane inclined at an angle of 30° with magnetic meridian is :
- (A) $\tan^{-1} \frac{1}{2}$ (B) $\tan^{-1} 2$ (C) $\tan^{-1} \frac{2}{3}$ (D) None of these

5. A coil in the shape of an equilateral triangle of side 0.02m is suspended from a vertex such that it is hanging in a vertical plane between the pole pieces of a permanent magnet producing a horizontal magnetic field of $5 \times 10^{-2} T$. Find the torque acting on the coil when a current of 0.1 ampere is passed through it and the magnetic field is parallel to its plane.
 (A) $8.65 \times 10^{-7} Nm$ (B) $8.65 \times 10^7 Nm$ (C) $8.68 \times 10^{-7} Nm$ (D) $8.68 \times 10^7 Nm$
6. 4.4 keV radiations belong to the spectrum portion of :
 (A) radio waves (B) microwaves (C) X-ray (D) infrared rays
7. A planoconvex lens behaves like a concave mirror of 30 cm focal length on silvering the plane surface of the lens, but on polishing by silver on convex surface, it acts as a concave mirror of 10 cm focal length. What will be the refractive index of the material of the lens?
 (A) 1 (B) 1.25 (C) 1.33 (D) 1.5
8. A man wants to see two poles separately situated at 11 km. The minimum approximate distance between the poles is : {Resolution limit of eye is $(1/60)^\circ$ }
 (A) 1 m (B) 3 m (C) 0.5 m (D) 5 m
9. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the slits. If the screen is moved by $5 \times 10^{-2} m$ towards the slits, the change in fringe width is $3 \times 10^{-5} m$. If the distance between the slits is $10^{-3} m$. Calculate the wavelength of light used.
 (A) $6 \times 10^{-7} m$ (B) $7 \times 10^{-7} m$ (C) $5 \times 10^{-7} m$ (D) $6 \times 10^{-6} m$
10. In which of the following circuits the current is maximum just after the switch S is closed :

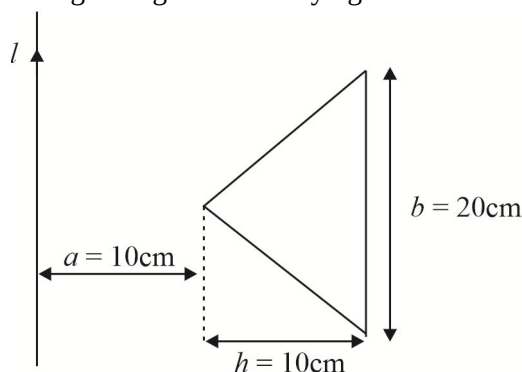


- (A) (i) (B) (ii) (C) (iii) (D) Both (ii) and (iii)
11. A circuit draws a power of 550W from a source of 220V, 50Hz. The power factor of the circuit is 0.8 and the current lags in phase behind the potential difference. To make the power factor of circuit 1.0, the capacitance required to be connected with it, will be :
 (A) $0.75 \mu F$ (B) $75 \mu F$ (C) $7.5 \mu F$ (D) $750 \mu F$
12. A ray of light strikes a plane mirror M at an angle of 45° as shown in the figure. After reflection, the ray passes through a prism of refractive index 1.5 whose apex angle is 4° . The total angle through which the ray is deviated is :



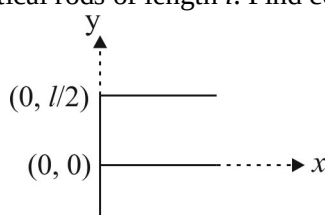
- (A) 90° (B) 91° (C) 92° (D) 93°

13. A silver sphere (work function 4.6eV) is suspended in a vacuum chamber by an insulating thread. Ultraviolet light of wavelength $0.2\mu\text{m}$ strikes on the sphere. The maximum electric potential of the sphere will be ($hc = 12400\text{ eV}\text{\AA}$):
 (A) 4.6V (B) 6.2V (C) 1.6V (D) 3.2V
14. Rain appears to be falling at 53° with the vertical to a man standing on the ground. When he starts running at 10 km/hr , rain again appears to be falling at 53° with the vertical. Find the speed (in km/hr) of rain wrt ground.
 (A) 10 (B) 5 (C) $\frac{25}{4}$ (D) $\frac{25}{3}$
15. A triangular loop lie near a straight long current carrying wire as shown in the figure.



Coefficient of mutual induction of coil with respect to the wire is :

- (A) $1.22 \times 10^{-8} H$ (B) $2.55 \times 10^{-8} H$ (C) $3.77 \times 10^{-8} H$ (D) $5.66 \times 10^{-8} H$
16. Letter F is formed from three identical rods of length l . Find coordinates of its centre of mass.



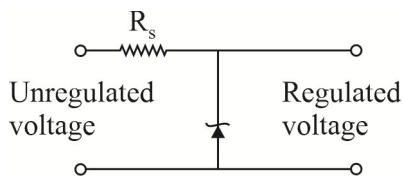
- (A) $\left(\frac{l}{5}, \frac{l}{5}\right)$ (B) $\left(\frac{l}{3}, \frac{l}{6}\right)$ (C) $\left(\frac{l}{6}, \frac{l}{3}\right)$ (D) $\left(\frac{l}{2}, \frac{l}{5}\right)$
17. A sample of H atoms in certain excited state emits 10 different wavelengths in de-excitation to ground state. The maximum wavelength out of these wavelengths is:
 (A) $4.05 \times 10^{-6} m$ (B) $4.05 \times 10^{-4} m$ (C) $4.5 \times 10^{-6} m$ (D) $4.5 \times 10^{-4} m$
18. A stone is dropped from the top of a tall cliff and n seconds later another stone is thrown vertically downwards with a velocity u . Then the second stone over takes the first, below the top of the cliff at a distance given by :

- (A) $\frac{g}{2} \left[\frac{n \left(\frac{gn}{2} - u \right)}{gn - u} \right]^2$ (B) $\frac{g}{2} \left[\frac{n \left(gn - \frac{u}{2} \right)}{gn - u} \right]^2$
- (C) $\frac{g}{2} \left[\frac{n \left(gn + \frac{u}{2} \right)}{gn - \frac{u}{2}} \right]^2$ (D) $\frac{g}{5} \left[\frac{gn - u}{gn - \frac{u}{2}} \right]^2$

19. Three planets of same density and with radii R_1, R_2 and R_3 , such that $R_1 = 2R_2 = 3R_3$ have gravitation fields on the surface g_1, g_2, g_3 and escape velocities v_1, v_2, v_3 respectively. Then :

(A) $\frac{g_1}{g_2} = \frac{1}{2}$ (B) $\frac{g_1}{g_3} = 3$ (C) $\frac{v_1}{v_2} = 4$ (D) $\frac{v_1}{v_3} = \frac{1}{3}$

20. A Zener of power rating 1W is to be used as a voltage regulator. If zener has a breakdown of 5V and it has to regulate voltage which fluctuated between 3V and 7V, what should be the value of R_s for safe operation (as shown in Figure)?



- (A) 20Ω (B) 5Ω (C) 10Ω (D) 30Ω

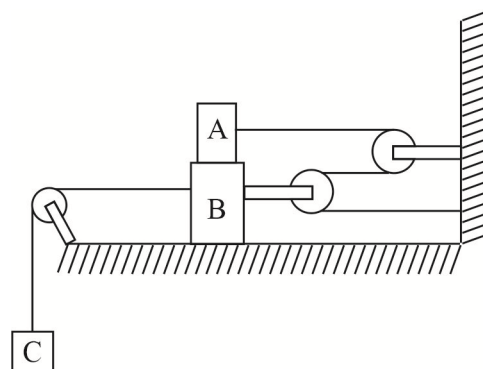
SPACE FOR ROUGH WORK

SECTION-2

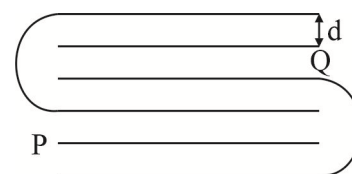
Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 questions have to be attempted.

The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

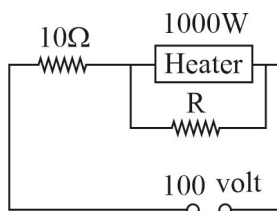
21. For the arrangement shown in figure, mass of A is 50 kg, mass of B is 70 kg, coefficient of static friction of all plane surfaces is $\mu_s = 0.3$. If the largest value of mass C (in kg), so that blocks A and B remain at rest) is M. Find $\frac{M}{9}$. (neglect friction in the pulleys).
[$g = 10 \text{ m/s}^2$]



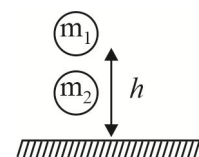
22. Six identical parallel metallic large plates are located in air at equal distance d to neighbouring plates. The area of each plate is A . Some of the plates are connected by conducting wires to each other. If the capacitance of the system of plates between points P and Q in pF is x . Find $\frac{x}{5}$.
(Take $A = 0.05 \text{ m}^2$, $d = 17.7 \text{ mm}$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$)



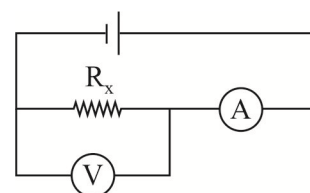
23. A 1000 watt heater is designed to be operated on a 100 volt line. It is connected to two resistances, 10Ω and R , as shown in figure. If it now gives a power of 62.5 watt. What is the value of R ?



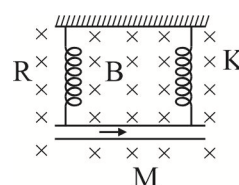
24. Two balls having masses m_1 and m_2 , having very little separation between them are dropped simultaneously from a certain height. All the collisions which take place are elastic. m_2 collides with ground and then rebounds and on rebounding collides with m_1 . Determine the ratio (m_2/m_1) so that m_2 delivers maximum percentage of its energy to m_1 .



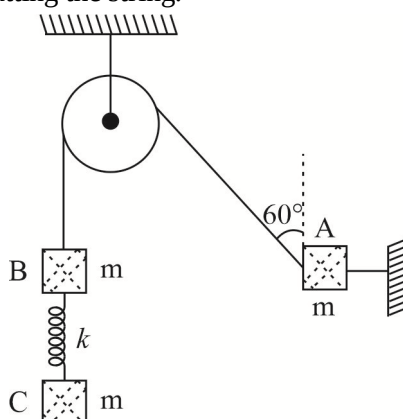
25. To measure the resistance of the resistor R_x , we use a circuit shown in figure. Ammeter shows current $I = 2.0 \text{ A}$ and voltmeter shows voltage $V = 120 \text{ volts}$. It voltmeter resistance is $R_V = 960\Omega$, what is the error in R_X (in ohm) if we use the approximate formula $R_X = \frac{V}{I}$?



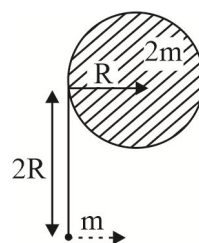
26. A metal rod of mass 10 gm and length 25cm is suspended on two springs as shown in the figure. The springs are extended by 4cm. When a 20 ampere current passes through the rod it rises by 1cm. The magnetic field is $x \times 10^{-2} T$ ($g = 10 m/s^2$). Find the value of $2x$.



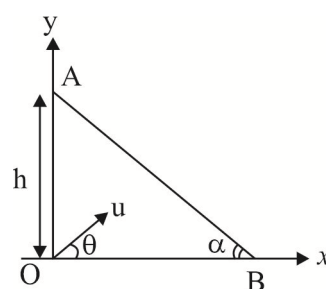
27. In the adjacent figure the blocks A, B and C are in equilibrium. The spring of stiffness k is light and vertical. The horizontal string joining the block A to a fixed wall is cut. If acceleration of B is $\frac{ng}{4}$, then find the value of n , just after cutting the string.



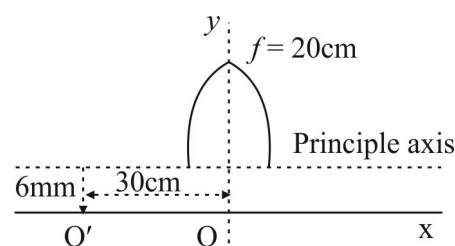
28. A uniform circular disc of mass $2m$ and radius R placed freely on a horizontal smooth surface as shown in the figure. A particle of mass m is connected to the circumference of the disc with a massless string. Now an impulse J is applied on the particle in the directions shown by dotted line. The acceleration (in m/s^2) of centre of mass of the disc just after application of impulse is (If $J = 10 N - sec$, $m = \sqrt{10} kg$ and $R = 25 cm$).



29. In the adjacent figure AB is an inclined wall making an angle $\alpha = 30^\circ$ with the horizontal (x -axis). AB crosses the y -axis at a height $h = \frac{10N}{3} m$. A small stone is projected from the origin 'O' with the speed $u = 10 m/s$ at angle 60° with the horizontal. Find the value of N , if the stone just grazes the wall. ($g = 10 m/s^2$).



30. A thin convex lens is cut along its principal axis and one half of the lens is placed as shown in the figure. The value of y coordinate is $9n$ (in mm), of the image with respect to the origin O of the given coordinate axis. The focal length of the lens is 20cm. Find the value of n . {Object is kept at O' }



SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**120 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Match the entries in column I to entries in column II and select correct alternative provided:

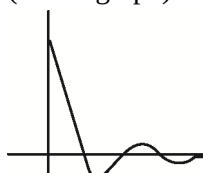
Column I

(Radial functions)

(A) Ψ_{3s} (B) Ψ_{4s} (C) Ψ_{3d} (D) Ψ_{3p} **Column II**

(R vs r graph)

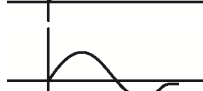
(u)



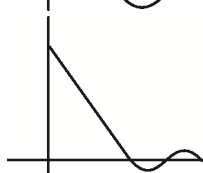
(v)



(w)



(x)



(A) A – w, B – v, C – u, D – x

(B) A – x, B – u, C – v, D – w

(C) A – u, B – x, C – v, D – w

(D) A – v, B – u, C – w, D – x

2. For the fuel cell reaction $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{H}_2\text{O}(\text{l})$; $\Delta_f H_{298}^\circ(\text{H}_2\text{O}, \text{l}) = -285.5 \text{ kJ/mol}$

Find the value of ΔS_{298}° and ΔG_{298}° for the given fuel cell reaction?

Given : $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O}(\text{l})$ $E^\circ = 1.23\text{V}$

(A) -0.322 kJ/K , -474.78 kJ (B) -0.635 kJ/K , -463.78 kJ (C) 3.51 kJ/K , -463.78 kJ (D) -0.322 J/K , -474.78 kJ

3. How many moles of NH_3 must be added to 6.0 litre of 0.80 M, AgNO_3 in order to reduce the Ag^+ concentration to $5 \times 10^{-8} \text{ M}$. K_f of $[\text{Ag}(\text{NH}_3)_2]^+ = 10^8$

(A) 4

(B) 2

(C) 12

(D) 10

4. Decomposition reaction $3\text{A}(\text{g}) \longrightarrow 2\text{B}(\text{g}) + 2\text{C}(\text{g})$ follows first order kinetics. Starting with pure 'A' pressure developed after 20 min and infinite time are 3.5 and 4 atm respectively. What is the time taken for 25% decomposition of A?

(A) 10 Min

(B) 20 Min

(C) 40 Min

(D) 5 Min

5. 1.0 molar aqueous solution of an electrolyte A_2B_3 is 70% ionized. The boiling point of the solution at 1 atm is: ($K_{b(\text{H}_2\text{O})} = 0.5 \text{ K kg mol}^{-1}$)

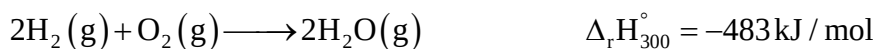
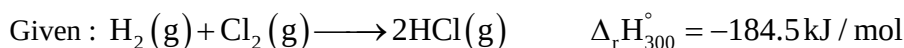
(A) 274.76 K

(B) 374.9 K

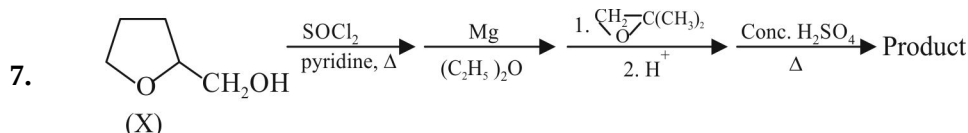
(C) 376.4 K

(D) 374.76 K

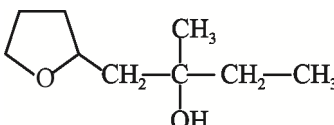
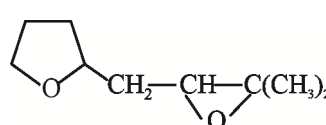
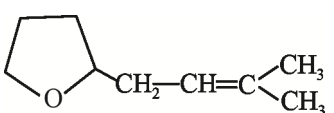
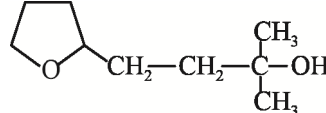
6. Find $\Delta_r H^\circ$ for the reaction $4\text{HCl}(\text{g}) + \text{O}_2(\text{g}) \longrightarrow 2\text{Cl}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ at 300 K. Assume all gases are ideal.

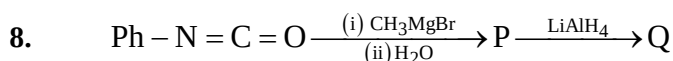


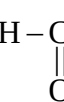
- (A) 111.5 kJ/mole (B) -109.01 kJ/mole
(C) -111.5 kJ/mole (D) -114 kJ/mol



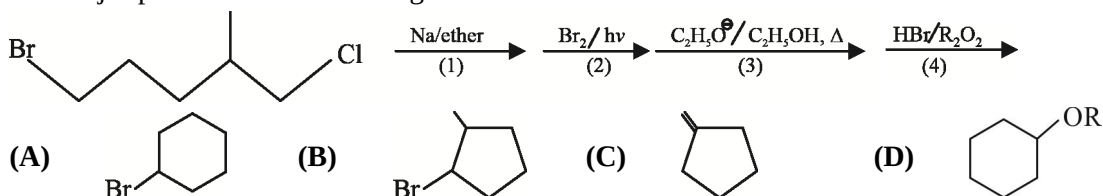
Product of the reaction is :

- (A)  (B) 
(C)  (D) 

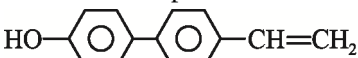
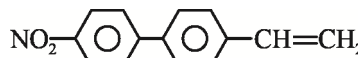
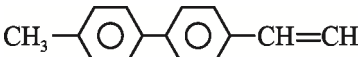
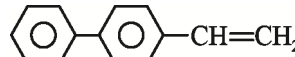


- (A) P is  and Q is $\text{Ph}-\text{NH}-\text{CH}_3$
(B) P is $\text{Ph}-\text{NH}-\text{CH}_3$ and Q is NH_3^{**}
(C) P is  and Q is $\text{Ph}-\text{NH}-\text{CH}_2-\text{CH}_3$
(D) P is  and Q is 

9. The major product of the following reaction is:



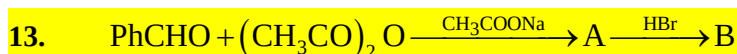
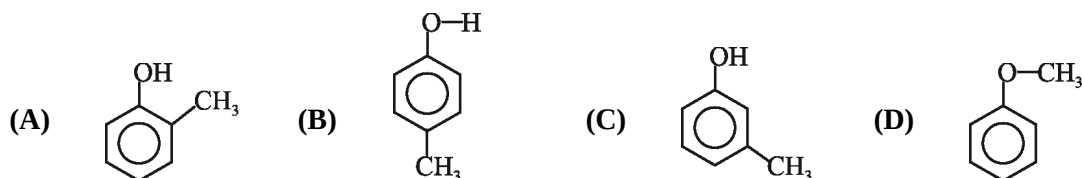
10. Slowest rate of electrophilic addition will take place in

- (A)  (B) 
(C)  (D) 

11. Among the following, the species that is both paramagnetic and coloured is:

- (A) $[\text{MnO}_4]^{2-}$ (B) $[\text{TiCl}_6]^{2-}$ (C) $[\text{VO}_4]^{3-}$ (D) CrO_2Cl_2

12. Compound Y, $\text{C}_7\text{H}_8\text{O}$ is insoluble in water, dil HCl and aqueous NaHCO_3 . It dissolves in dilute NaOH. When Y is treated with bromine water it is converted rapidly into a compound of formula $\text{C}_7\text{H}_5\text{OBr}_3$. Identify the structure of Y.



The product B is :

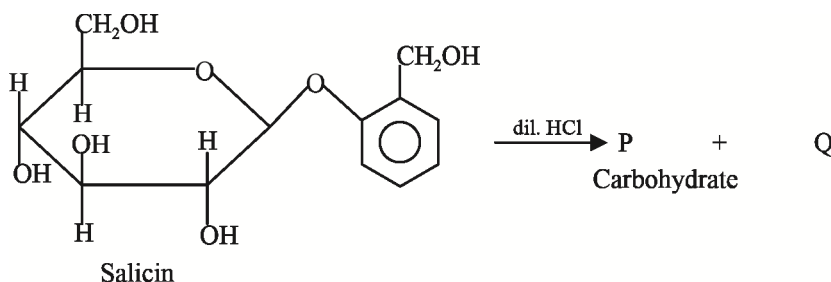
- (A) $\text{PhCH}=\text{CHCH}_2\text{Br}$ (B) $\text{PhCH}(\text{Br})\text{CH}_2\text{COOH}$
(C) $\text{PhCH}_2\text{CH}(\text{Br})\text{COOH}$ (D) $\text{PhCH}=\text{CHCOBr}$

14. Which of the following statements are not true for antibiotics ?

- I. They are produced wholly or partly by chemical synthesis
II. In low concentrations, they inhibit the growth or destroy microorganisms by intervening in their metabolic processes
III. Even in low doses, they are highly toxic for humans
IV. They are products of microorganisms as purely synthetic compounds can never have antibacterial activity

- (A) III, IV (B) I, II (C) II, IV (D) I, III

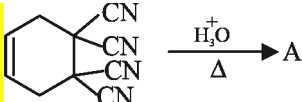
15. Salicin (structure given below) is a glycoside, found in the bark of willow tree, used in relieving pain. Observe the following reaction of salicin

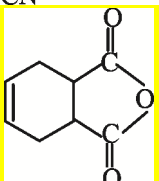
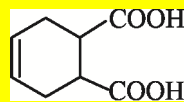


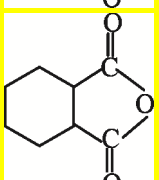
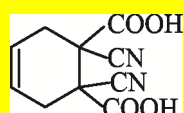
The correct statement(s) is (are) :

- I. P is D- glucose II. Q is 2-hydroxybenzylalcohol
III. P is L-glucose IV. The above reaction occurs through a carbocation
(A) I, II, III (B) I, II, IV (C) I, IV (D) I, II

16. Main ore of lead is galena. This is mined and separated from other minerals by froth flotation. There are two methods of extracting the lead.
- First method involves the roasting of ore followed by reduction with coke or CO.
 - Second method involves the partial roasting of ore followed by self reduction.
- Which of the following statements is correct about the extraction of lead from galena?
- Galena is a sulphide ore and therefore, it is concentrated by froth flotation process.
 - Self reduction takes place in absence of air.
 - Complete roasting of galena gives PbO which is then reduced by coke or CO to give metallic lead
 - FeSiO₃ is obtained as slag.
- (A) P, Q, R (B) R, S (C) P, Q, R, S (D) None

17.  Product (A) of the reaction is :

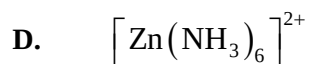
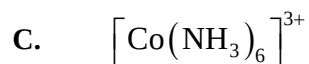
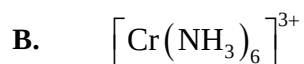
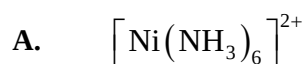
(A)  (B) 

(C)  (D) 

18. **Column –I** **Column –II**
- | | |
|---|---|
| A. $\text{XeF}_4(\text{s}) + \text{H}_2\text{O}(\text{l}) \longrightarrow$ | (P). Show disproportionation reaction. |
| B. $\text{P}_4(\text{s}) + \text{OH}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow$ | (Q). One of the products is a monobasic Lewis acid. |
| C. $\text{B}_2\text{H}_6(\text{g}) + \text{H}_2\text{O}(\text{l}) \longrightarrow$ | (R). One of the products is a gas. |
| D. $\text{Br}_2(\text{g}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow$ | (S). One of the products has a trigonal pyramidal molecular structure |
| | (T). One of the gaseous products is used in Holme's signal in sea. |
- (A) A → p, t; B → p, r, s, t; C – q, r; D – p, r, s
- (B) A → p, r, s; B → p, r, s, t; C – q, r; D – p, r, s
- (C) A → p, r, s; B → s, C – t, q, r; D – p, r, s
- (D) A → p, r, s; B → p, r, s, t; C – q, r; D – t
19. Which of the following statements is /are true?
- Cu²⁺ salts form soluble complex with excess KCN
 - Cu²⁺ salts form soluble complex with aqueous ammonia
 - Cu²⁺ salts form soluble complex with KI.
 - A piece of iron or zinc when placed in Cu²⁺ salt solution, precipitates copper.
- (A) I, III, IV (B) I, II, III (C) I, II, IV (D) None

20. Match the complex ions listed in column –I with the characteristics listed in column –II using the codes gives below.

Column I



Column II

(P). d^2sp^3 and diamagnetic

(q). sp^3d^2 and diamagnetic

(r). sp^3d^2 and two unpaired electrons

(s). d^2sp^3 and three unpaired electrons.

	A	B	C	D
(A)	s	r	q	p
(B)	r	s	p	q
(C)	r	s	q	p
(D)	p	q	r	s

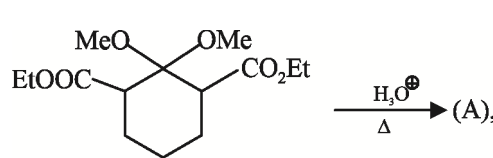
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SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 questions have to be attempted.

The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

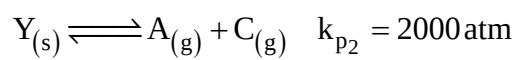
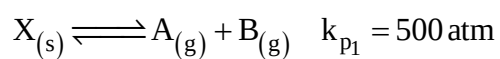
21. Total number of Hexagonal faces in a truncated octahedron = x
Total number of Hexagonal faces in a truncated tetrahedron = y
Then $x + y$ is:
22. The pressure exerted by 12 g of ideal gas at temperature $t^\circ\text{C}$ in a vessel of volume V litre is one atm. When the temperature is increased by 10 degree at the same volume, the pressure is increased by 10% . Calculate the temperature t in Kelvin. (Molecular weight of the gas = 120)
23. 50 ml of a colloidal solution is completely precipitated by addition of 0.5 ml of 1 M NaCl solution. What will be the coagulation value.
24. X = Sum of antibonding π electrons (π^* electrons) in species O_2 , O_2^- and O_2^{2-} .
Y = The number of species, among the following species, where central atoms have two lone pairs of electrons
- | | | |
|------------------|------------------|--------------------------|
| XeF_4 | XeF_5^- | F_2SeO_2 |
| XeF_3^+ | XeOF_4 | ClOF_3 |
| ICl_4^- | SCl_2 | OSF_4 |
- Find the value of $X + Y$.
25. The total number of carbonate ores among the following is.
(i) Argentite (ii) Calamine (iii) Malachite (iv) Siderite (v) Galena
(vi) Magnetite (vii) Dolomite (viii) Cassiterite (ix) Bauxite
26. If Phosphorus acid, Tetrathionic acid and Pyrophosphoric acid have number of acidic hydrogen per molecule respectively as x, y and z, then find the value of $x + y + z$
27. How many Cr – O bonds are equivalent in dichromate dianion?
28. The possible number of stereoisomers of the product of following reaction would be:

$$\text{Ph} - \text{CH} = \text{CH} - \underset{\text{CH}_3}{\text{CH}} - \text{CHO} \xrightarrow{\text{NH}_2\text{OH}}$$
29. 

$$\xrightarrow[\Delta]{\text{H}_3\text{O}^+} (\text{A}),$$

The molecular weight of the organic product (A) is:

30. $X_{(s)}$ and $Y_{(s)}$ are taken in a closed vessel and decompose as follows:



Find the total pressure developed in the vessel.

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PART - III : MATHEMATICS**120 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Let $z = a + ib$ be a complex number such that $a, b \in \mathbb{Q}$, $|z| = 1$, $|z^{2n} - 1| = x$. Then $(-1)^x$ is :
 (A) positive real number (B) negative real number
 (C) a complex number not necessarily real (D) can't decide
- The sum of infinite series :
 $\frac{3}{4} + \frac{5}{36} + \frac{7}{144} + \frac{9}{400} + \frac{11}{900} + \dots \infty$ is
 (A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) 2
- The number of permutations of 25 dissimilar things taken more than 15 at a time when repetitions are allowed is (not exceeding 25):
 (A) $\frac{25}{24}(25^{25} - 25^{15})$ (B) $\frac{25}{24}(25^{25} - 25^{10})$
 (C) $\frac{25}{24}(25^{25} + 25^{15})$ (D) $\frac{25}{24}(25^{25} + 25^{10})$
- If $\log_{1/10}(x^2 + x) > \log_{1/2}(x^3 - x) + \log_2(x - 1)$, then :
 (A) $x \in (1, \infty)$ (B) $x \in \left(\frac{\sqrt{5}-1}{2}, 1\right)$
 (C) $x \in \left(\frac{-\sqrt{5}-1}{2}, \frac{\sqrt{5}-1}{2}\right)$ (D) $x \in \left(-\infty, \frac{-\sqrt{5}-1}{2}\right)$
- A triangle $\triangle ABC$ has sides of lengths $|z_1|, |z_2|$ and $|z_3|$ such that $|z_1|(|z_1| - |z_2|) + |z_2|(|z_2| - |z_3|) + |z_3|(|z_3| - |z_1|) = 0$. Then which of the following can be a possible pair of values of $(|z_1|, R)$ (R = circumradius of $\triangle ABC$)?
 (A) $(1, \sqrt{2})$ (B) $(\sqrt{2}, 1)$ (C) $(1, \sqrt{3})$ (D) $(\sqrt{3}, 1)$
- The value of ${}^{200}C_0 - {}^{200}C_1 + {}^{200}C_2 - {}^{200}C_3 + \dots + {}^{200}C_{148}$ is equal to :
 (A) ${}^{199}C_{148}$ (B) ${}^{198}C_{51}$ (C) ${}^{200}C_{148}$ (D) ${}^{189}C_{148}$
- The number of permutations which can be formed out of the letters of the word "SERIES" taking three letters together, is :
 (A) 30 (B) 60 (C) 56 (D) 42
- If α and β are roots of equation $x^2 - 7x - 1 = 0$, then the value of $\frac{(\alpha^{10} - \beta^{10}) - (\alpha^8 - \beta^8)}{\alpha^{(\alpha+\beta+2)} - \beta^{(\alpha+\beta+2)}}$ is equal to :
 (A) 1 (B) 0 (C) 7 (D) $\frac{1}{7}$

9. The value of $\lim_{x \rightarrow 0^+} \left(\min. (t^2 - 6(t+x) + 12) + \left\lceil \frac{\log_e(1+2x)}{x} \right\rceil \right)$, where $t \in R$ and $[y]$ denotes greatest integer less than or equal to y , is :
- (A) 4 (B) 3 (C) 6 (D) 9
10. Let $f(x) = \begin{cases} \max. \{e^x, e^{-x}, 2\} & x \leq 0 \\ \min. \{e^x, e^{-x}, 2\} & x > 0 \end{cases}$
- Which of the following statements is **NOT** true?
- (A) $f(x)$ is discontinuous at $x = 0$
 (B) $f(x)$ is non-derivable at exactly two points
 (C) $f(x)$ has non-removable type of discontinuity at $x = 0$ with jump of discontinuity equal to 2.
 (D) $f(x)$ is continuous but non-derivable at $x = \log_e \frac{1}{2}$
11. Let $f(x) = \sin \frac{\{x\}}{a} + \cos \frac{\{x\}}{a}$, where $a > 0$ and $\{\cdot\}$ denotes the fraction part function. Then the set of values of a for which f can attain its maximum values is:
- (A) $\left(0, \frac{4}{\pi}\right)$ (B) $\left(\frac{4}{\pi}, \infty\right)$ (C) $(0, \infty)$ (D) R
12. If $0 < x < 1$ and $\frac{d}{dx} \left\{ \cos^{-1} \left(\frac{1-x}{1+x} \right) \right\} = \frac{1}{f(x) \{1 + f^2(x)\}}$, then $\int \frac{d\{f^2(x)\}}{f(x) + f^2(x)} =$
- (A) $2 \ln |1 + f^2(x)| + c$ (B) $2 \ln |1 + f(x)| + c$
 (C) $\ln |1 + f(x)| + c$ (D) $\ln |1 + f^2(x)| + c$
13. If area bounded by $y = f(x)$, y -axis and the line $2y = \pi(x+1)$, where $f(x) = \sin^{-1} x + \cos^{-1} x + \tan^{-1} x + \tan^{-1} \frac{1}{x}$ is :
- (A) $\pi/4$ (B) $\pi/2$ (C) π (D) 2π
14. Direction cosines of normal to plane containing the lines $L_1 : \frac{x-1}{2} = \frac{y-1}{1}$ and $z = 2$ and $L_2 : \frac{x-2}{1} = \frac{y-3}{2} = \frac{z-2}{3}$ are :
- (A) $\left\{ \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\}$ (B) $\left\{ \frac{-1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{-1}{\sqrt{6}} \right\}$
 (C) $\left\{ \frac{1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right\}$ (D) $\left\{ \frac{-1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\}$
15. The solution of the differential equation $y - x \frac{dy}{dx} = a \left(y^2 + \frac{dy}{dx} \right)$ is :
- (A) $x = c(1+ay)(x+a)$ (B) $y = c(1+ay)(x-a)$
 (C) $y = c(1-ay)(x+a)$ (D) $y = c(1-ay)(ax+1)$

16. A bag contains some white and some black balls, all combinations of balls being equally likely. The total number of balls in the bag is 10. If 3 balls are drawn at random without replacement and all of them are found to be black, the probability that the bag contains 1 white and 9 black balls is :
 (A) 14/55 (B) 12/55 (C) 2/11 (D) 22/55
17. If $x \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\sin^{-1}(\cos x) + \cos^{-1}(\sin x)}{\tan^{-1}(\cot x) + \cot^{-1}(\tan x)} =$
 (A) $\pi/4$ (B) $\pi/3$ (C) $\pi/2$ (D) 1
18. If $f(x)$ is a differentiable function such that $f'(1) = 4$ and $f'(4) = \frac{1}{2}$, then value of $\lim_{x \rightarrow 0} \frac{f(x^2 + x + 1) - f(1)}{f(x^4 - x^2 + 2x + 4) - f(4)}$:
 (A) 8 (B) 16 (C) 4 (D) Does not exist
19. The value of $\int \frac{1 + (\sin x)^{2/3}}{1 + (\sin x)^{4/3}} d(\sqrt[3]{\sin x})$:
 (A) $\sin^{-1}(1 + \sqrt[3]{\sin x}) + c$ (B) $\sin^{-1}(\sqrt[3]{\sin x}) + c$
 (C) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt[3]{\sin^2 x} - 1}{\sqrt{2} \sqrt[3]{\sin x}} \right) + c$ (D) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt[3]{\sin x}}{\sqrt{2}} \right) + c$
20. The value of $\int_{1/3}^{2/3} \frac{\log_e x}{\log_e (x - x^2)} dx$ is :
 (A) 1/3 (B) 1/6 (C) 1/2 (D) 1

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21. $\sum_{r=0}^{18} {}^{20}C_r \cdot {}^{20}C_{r+2} = {}^n C_R$, where $R > 20$ then $\frac{n-R}{2} = \underline{\hspace{2cm}}$.
22. If the sum of the G.P. is 63.5 except for the first term, sum is 127 except the last one, and the sum is 31.5 except the first two terms, then the number of terms in G.P. is $\underline{\hspace{2cm}}$.
23. If $\int e^{\left(\frac{x^6+x^2-x-1}{x-1}\right)} (5x^4 + 4x^3 + 3x^2 + 2x + 2) dx = e^{g(x)} + C$ where C is constant of integration and $g(0) = 1$, then the value of $g(1)$ is $\underline{\hspace{2cm}}$.
24. Given $f(x) = \sin^3 x$ and $p(x)$ is a quadratic polynomial with leading coefficient unity. Then the value of $\int_0^{2\pi} p(x) f''(x) dx$ is $\underline{\hspace{2cm}}$.
25. Let $f: R \rightarrow R$ be a differentiable function satisfying $f(1) = 2$. If α, β are real numbers satisfying $\alpha^2 + \beta^2 = 1$, then $f(\alpha x) \cdot f(\beta x) = f(x)$ for all $x \in R$. Then the value of $f(0) + f'(0)$ is $\underline{\hspace{2cm}}$.
26. A bag contains 10 balls of which 2 are red and the remaining are either blue or black. If the probability of drawing 3 balls of the same color is $\frac{11}{120}$ and if the number of blue balls exceeds the number of black balls, the number of blue balls is $\underline{\hspace{2cm}}$.
27. For $n = 1, 2, \dots$, let $S_n = \sum_{k=0}^{3n} {}^{3n}C_k$ and $T_n = \sum_{k=0}^n {}^{3n}C_{3k}$, then the value of $|S_n - 3T_n| = \underline{\hspace{2cm}}$.
28. There are four different boxes A, B, C and D and four different balls α, β, γ and δ . The number of ways of filling the balls in the boxes such that each box contains one ball, ball α cannot go to the box A and ball β cannot go to the box B , is $\underline{\hspace{2cm}}$.
29. If $1 + \sum_{r=0}^{18} \{r(r+2)+1\} r! = k!$ then k is $\underline{\hspace{2cm}}$.
30. Let P and Q are two non-singular square matrices of order 3×3 such that $PQ = P$ and $QP = Q$. If $\det. (P + Q) \neq 0$, then the value of $\det. (P + Q) + \det. (P - Q)$ is $\underline{\hspace{2cm}}$.

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